

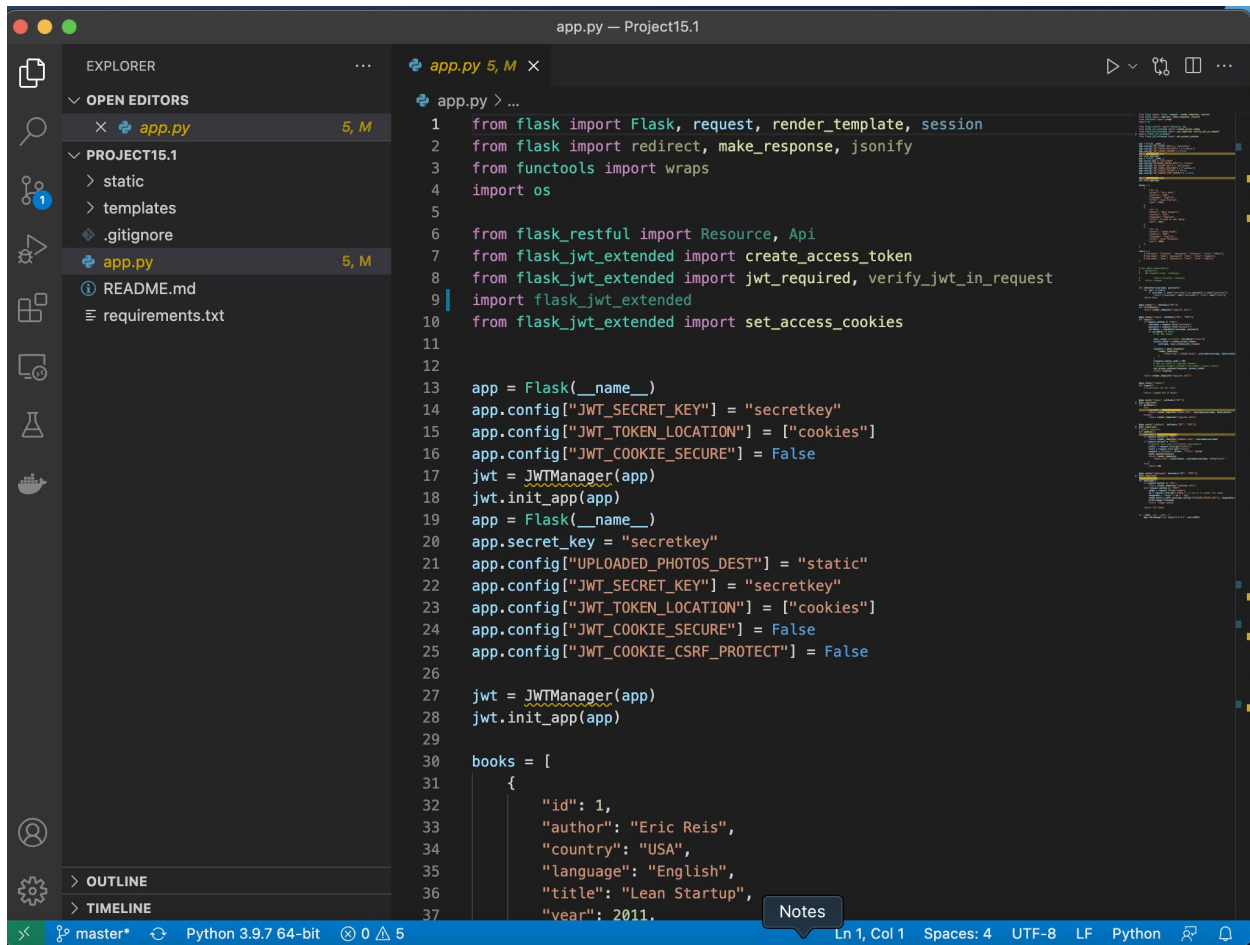
Project 15.1 Solution File

1.

```
Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.9/site-packages (from Flask>=0.8->flask-restful) (8.0.3)
Requirement already satisfied: itsdangerous>=2.0 in /usr/local/lib/python3.9/site-packages (from Flask>=0.8->flask-restful) (2.0.1)
Requirement already satisfied: Jinja2>=3.0 in /usr/local/lib/python3.9/site-packages (from Flask>=0.8->flask-restful) (3.0.2)
Requirement already satisfied: Werkzeug>=2.0 in /usr/local/lib/python3.9/site-packages (from Flask>=0.8->flask-restful) (2.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.9/site-packages (from Jinja2>=3.0->Flask>=0.8->flask-restful) (2.0.1)
Installing collected packages: aniso8601, flask-restful
  DEPRECATION: Configuring installation scheme with distutils config files is deprecated and will no longer work in the near future. If you are using a Homebrew or Linuxbrew Python, please see discussion at https://github.com/Homebrew/homebrew-core/issues/76621
  DEPRECATION: Configuring installation scheme with distutils config files is deprecated and will no longer work in the near future. If you are using a Homebrew or Linuxbrew Python, please see discussion at https://github.com/Homebrew/homebrew-core/issues/76621
Successfully installed aniso8601-9.0.1 flask-restful-0.3.9
WARNING: You are using pip version 21.2.4; however, version 21.3.1 is available.
You should consider upgrading via the '/usr/local/opt/python@3.9/bin/python3.9 -m pip install --upgrade pip' command.
> pip3 install flask-jwt-extended
```

```
Collecting PyJWT<3.0,>=2.0
  Downloading PyJWT-2.3.0-py3-none-any.whl (16 kB)
Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.9/site-packages (from Flask<3.0,>=1.0->flask-jwt-extended) (8.0.3)
Requirement already satisfied: itsdangerous>=2.0 in /usr/local/lib/python3.9/site-packages (from Flask<3.0,>=1.0->flask-jwt-extended) (2.0.1)
Requirement already satisfied: Jinja2>=3.0 in /usr/local/lib/python3.9/site-packages (from Flask<3.0,>=1.0->flask-jwt-extended) (3.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.9/site-packages (from Jinja2>=3.0->Flask<3.0,>=1.0->flask-jwt-extended) (2.0.1)
Installing collected packages: PyJWT, flask-jwt-extended
  DEPRECATION: Configuring installation scheme with distutils config files is deprecated and will no longer work in the near future. If you are using a Homebrew or Linuxbrew Python, please see discussion at https://github.com/Homebrew/homebrew-core/issues/76621
  DEPRECATION: Configuring installation scheme with distutils config files is deprecated and will no longer work in the near future. If you are using a Homebrew or Linuxbrew Python, please see discussion at https://github.com/Homebrew/homebrew-core/issues/76621
Successfully installed PyJWT-2.3.0 flask-jwt-extended-4.3.1
WARNING: You are using pip version 21.2.4; however, version 21.3.1 is available.
You should consider upgrading via the '/usr/local/opt/python@3.9/bin/python3.9 -m pip install --upgrade pip' command.
```

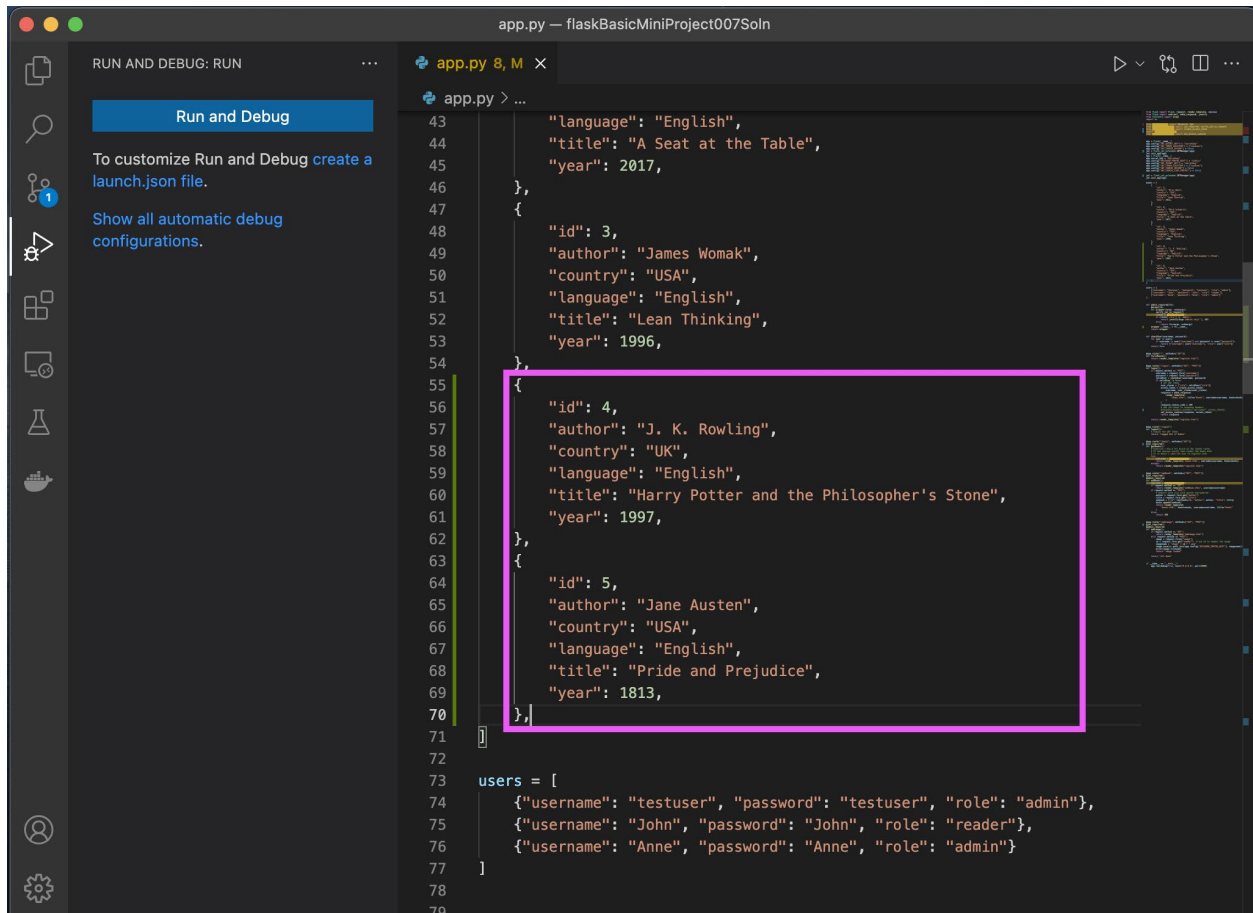
2.



The screenshot shows a VS Code editor window titled "app.py - Project15.1". The Explorer sidebar on the left shows the project structure for "PROJECT15.1", including files like "static", "templates", ".gitignore", "app.py", "README.md", and "requirements.txt". The main editor area displays the code for "app.py", which is a Flask application using JWT authentication. The code includes imports for Flask, request, render_template, session, redirect, make_response, jsonify, functools wraps, and os. It also imports Resource and Api from flask_restful, and create_access_token, jwt_required, verify_jwt_in_request, and set_access_cookies from flask_jwt_extended. The application configuration sets the secret key to "secretkey", the token location to "cookies", and disables cookie security. A JWTManager is initialized, and the app is configured with the uploaded photos destination as "static". A list of books is defined at the bottom of the code.

```
1 from flask import Flask, request, render_template, session
2 from flask import redirect, make_response, jsonify
3 from functools import wraps
4 import os
5
6 from flask_restful import Resource, Api
7 from flask_jwt_extended import create_access_token
8 from flask_jwt_extended import jwt_required, verify_jwt_in_request
9 import flask_jwt_extended
10 from flask_jwt_extended import set_access_cookies
11
12
13 app = Flask(__name__)
14 app.config["JWT_SECRET_KEY"] = "secretkey"
15 app.config["JWT_TOKEN_LOCATION"] = ["cookies"]
16 app.config["JWT_COOKIE_SECURE"] = False
17 jwt = JWTManager(app)
18 jwt.init_app(app)
19 app = Flask(__name__)
20 app.secret_key = "secretkey"
21 app.config["UPLOADED_PHOTOS_DEST"] = "static"
22 app.config["JWT_SECRET_KEY"] = "secretkey"
23 app.config["JWT_TOKEN_LOCATION"] = ["cookies"]
24 app.config["JWT_COOKIE_SECURE"] = False
25 app.config["JWT_COOKIE_CSRF_PROTECT"] = False
26
27 jwt = JWTManager(app)
28 jwt.init_app(app)
29
30 books = [
31     {
32         "id": 1,
33         "author": "Eric Reis",
34         "country": "USA",
35         "language": "English",
36         "title": "Lean Startup",
37         "year": 2011,
```

3.



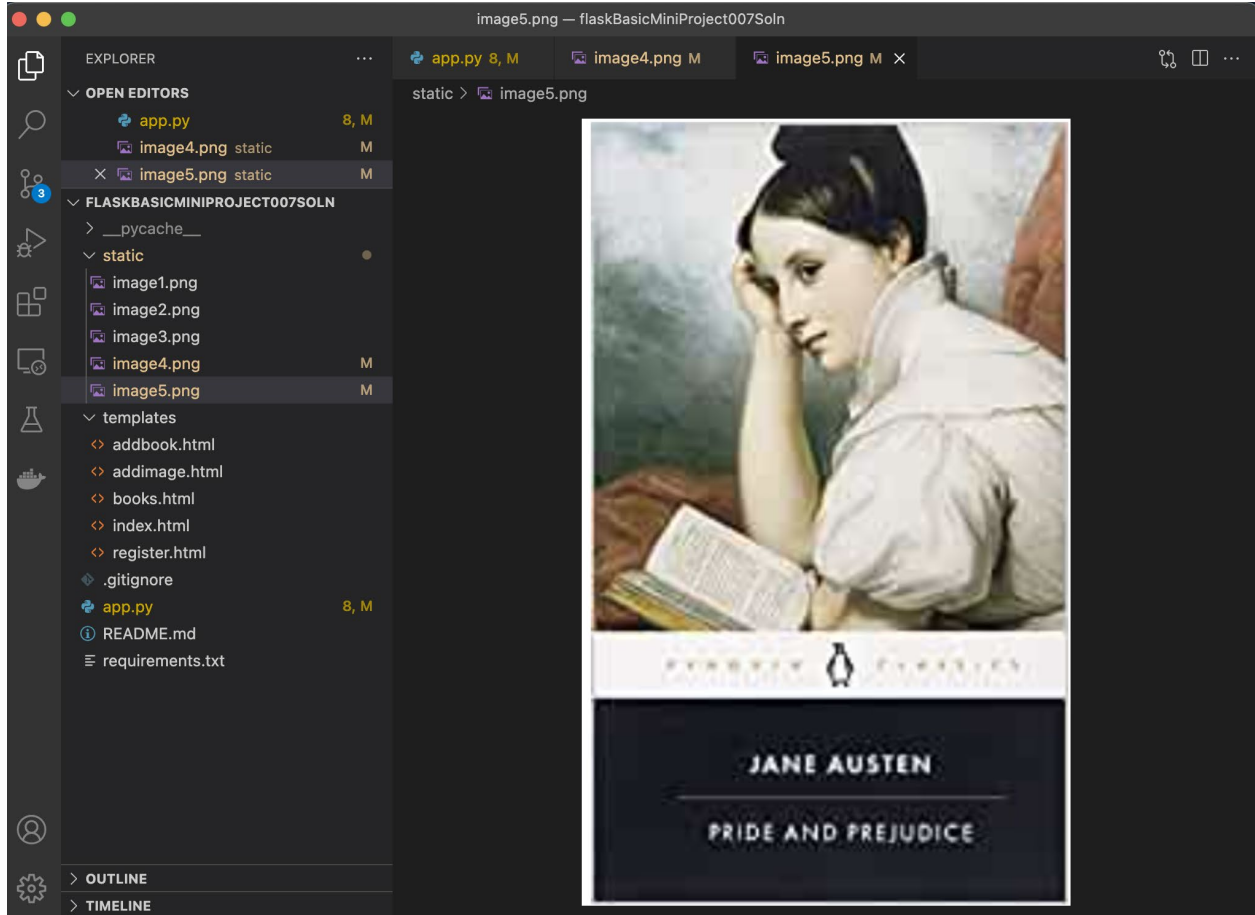
The screenshot shows a VS Code editor window titled "app.py — flaskBasicMiniProject007Soln". The editor is displaying a Python file named "app.py" with the following code:

```
43     "language": "English",
44     "title": "A Seat at the Table",
45     "year": 2017,
46 },
47 {
48     "id": 3,
49     "author": "James Womak",
50     "country": "USA",
51     "language": "English",
52     "title": "Lean Thinking",
53     "year": 1996,
54 },
55 {
56     "id": 4,
57     "author": "J. K. Rowling",
58     "country": "UK",
59     "language": "English",
60     "title": "Harry Potter and the Philosopher's Stone",
61     "year": 1997,
62 },
63 {
64     "id": 5,
65     "author": "Jane Austen",
66     "country": "USA",
67     "language": "English",
68     "title": "Pride and Prejudice",
69     "year": 1813,
70 },
71 ]
72
73 users = [
74     {"username": "testuser", "password": "testuser", "role": "admin"},
75     {"username": "John", "password": "John", "role": "reader"},
76     {"username": "Anne", "password": "Anne", "role": "admin"}
77 ]
78
79
```

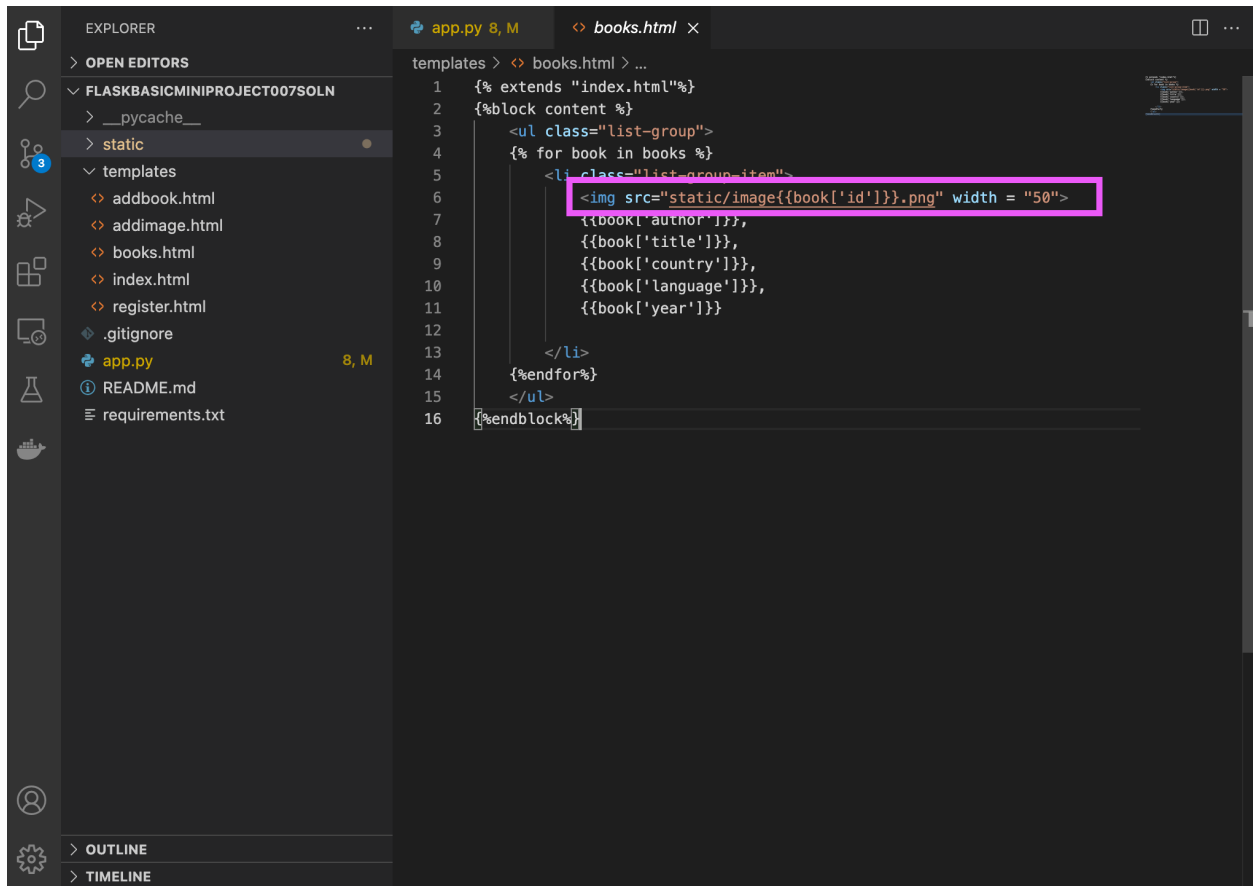
A pink box highlights the book object with id 4, author "J. K. Rowling", country "UK", language "English", title "Harry Potter and the Philosopher's Stone", and year 1997.

4.



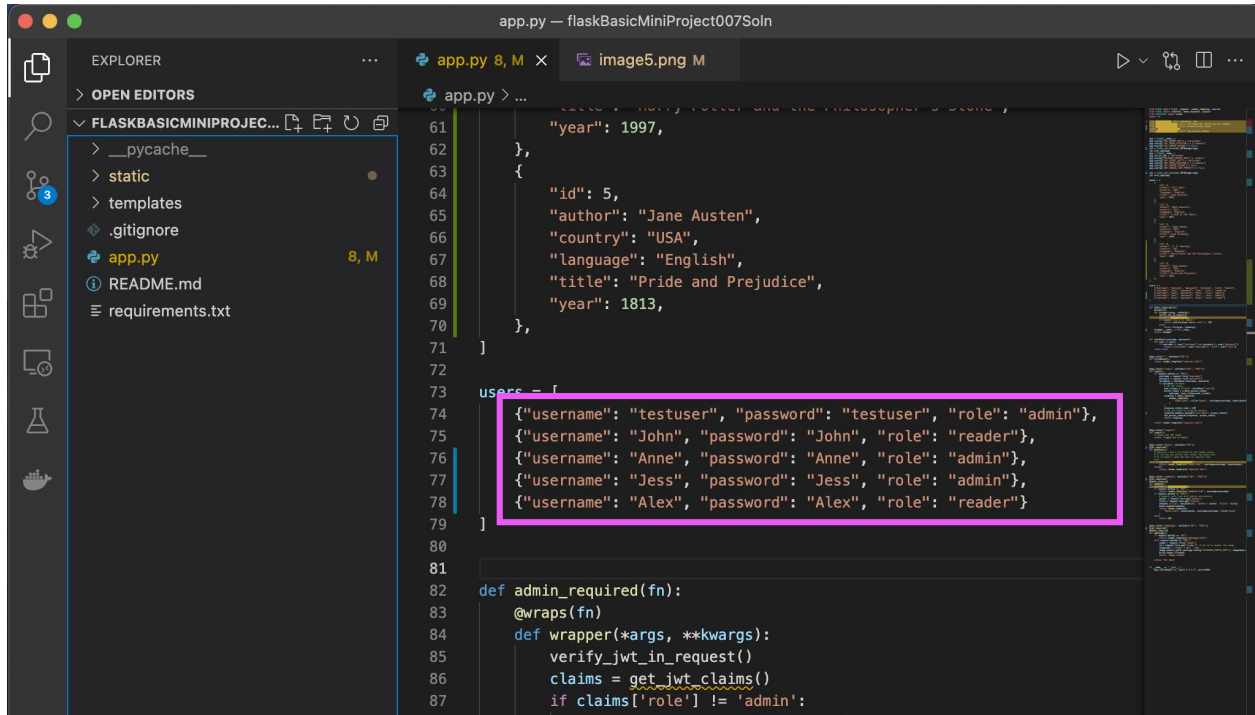


5.



```
1 {% extends "index.html"%}
2 {%block content %}
3
4 <ul class="list-group">
5   {% for book in books %}
6     <li class="list-group-item">
7       
8       {{book['author']}},
9       {{book['title']}},
10      {{book['country']}},
11      {{book['language']}},
12      {{book['year']}}
13     </li>
14   {%endfor%}
15 </ul>
16 {%endblock%}
```

6.

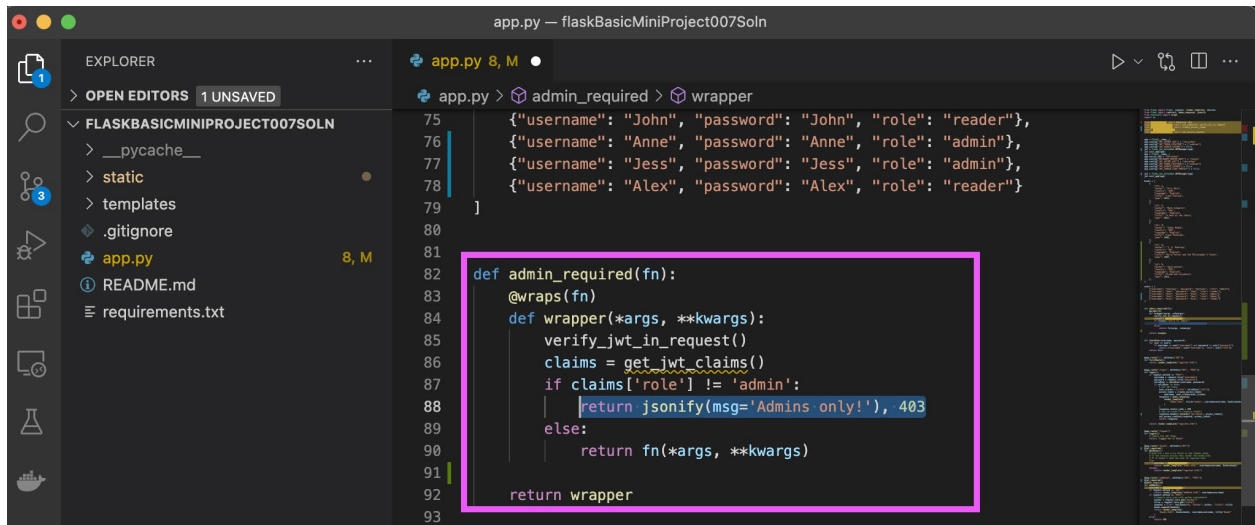


```

app.py — flaskBasicMiniProject007Soln
EXPLORER
  > OPEN EDITORS
  > FLASKBASICMINIPROJECT007SOLN
    > __pycache__
    > static
    > templates
    > .gitignore
    > app.py 8, M
    > README.md
    > requirements.txt
  > app.py 8, M
    61     "year": 1997,
    62   },
    63   {
    64     "id": 5,
    65     "author": "Jane Austen",
    66     "country": "USA",
    67     "language": "English",
    68     "title": "Pride and Prejudice",
    69     "year": 1813,
    70   },
    71 ]
    72
    73 users = [
    74   {"username": "testuser", "password": "testuser", "role": "admin"},
    75   {"username": "John", "password": "John", "role": "reader"},
    76   {"username": "Anne", "password": "Anne", "role": "admin"},
    77   {"username": "Jess", "password": "Jess", "role": "admin"},
    78   {"username": "Alex", "password": "Alex", "role": "reader"}
    79 ]
    80
    81
    82 def admin_required(fn):
    83     @wraps(fn)
    84     def wrapper(*args, **kwargs):
    85         verify_jwt_in_request()
    86         claims = get_jwt_claims()
    87         if claims['role'] != 'admin':

```

7.

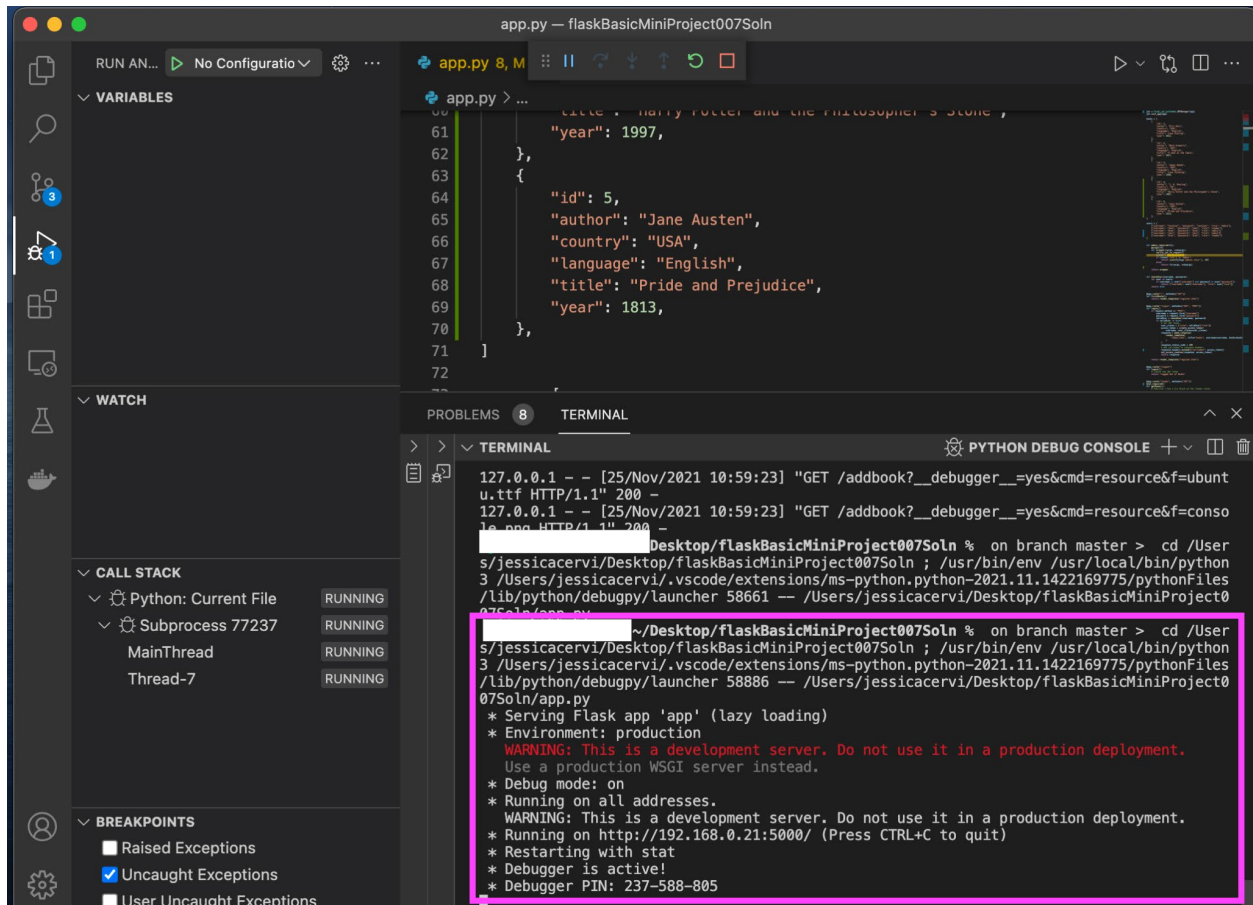


```

app.py — flaskBasicMiniProject007Soln
EXPLORER
  > OPEN EDITORS 1 UNSAVED
  > FLASKBASICMINIPROJECT007SOLN
    > __pycache__
    > static
    > templates
    > .gitignore
    > app.py 8, M
    > README.md
    > requirements.txt
  > app.py 8, M
    75   {"username": "John", "password": "John", "role": "reader"},
    76   {"username": "Anne", "password": "Anne", "role": "admin"},
    77   {"username": "Jess", "password": "Jess", "role": "admin"},
    78   {"username": "Alex", "password": "Alex", "role": "reader"}
    79 ]
    80
    81
    82 def admin_required(fn):
    83     @wraps(fn)
    84     def wrapper(*args, **kwargs):
    85         verify_jwt_in_request()
    86         claims = get_jwt_claims()
    87         if claims['role'] != 'admin':
    88             return jsonify(msg='Admins only!'), 403
    89         else:
    90             return fn(*args, **kwargs)
    91     return wrapper
    92
    93

```

8.



9.

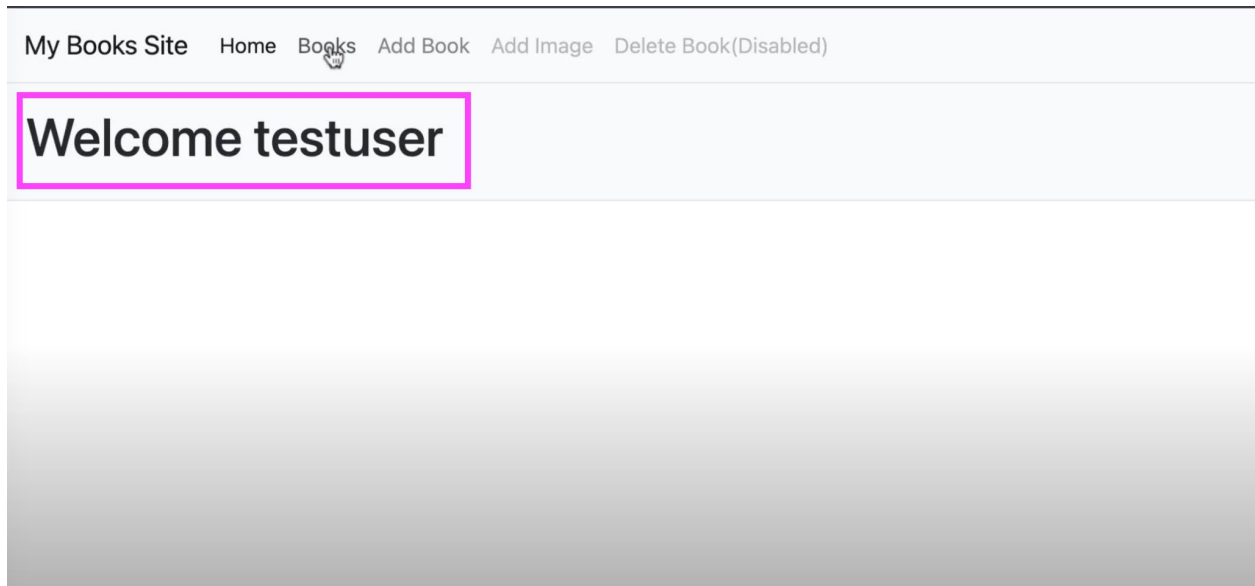
My Books Site
Home
Books
Add Book
Add Image
Delete Book(Disabled)

Use "testuser" for username and password

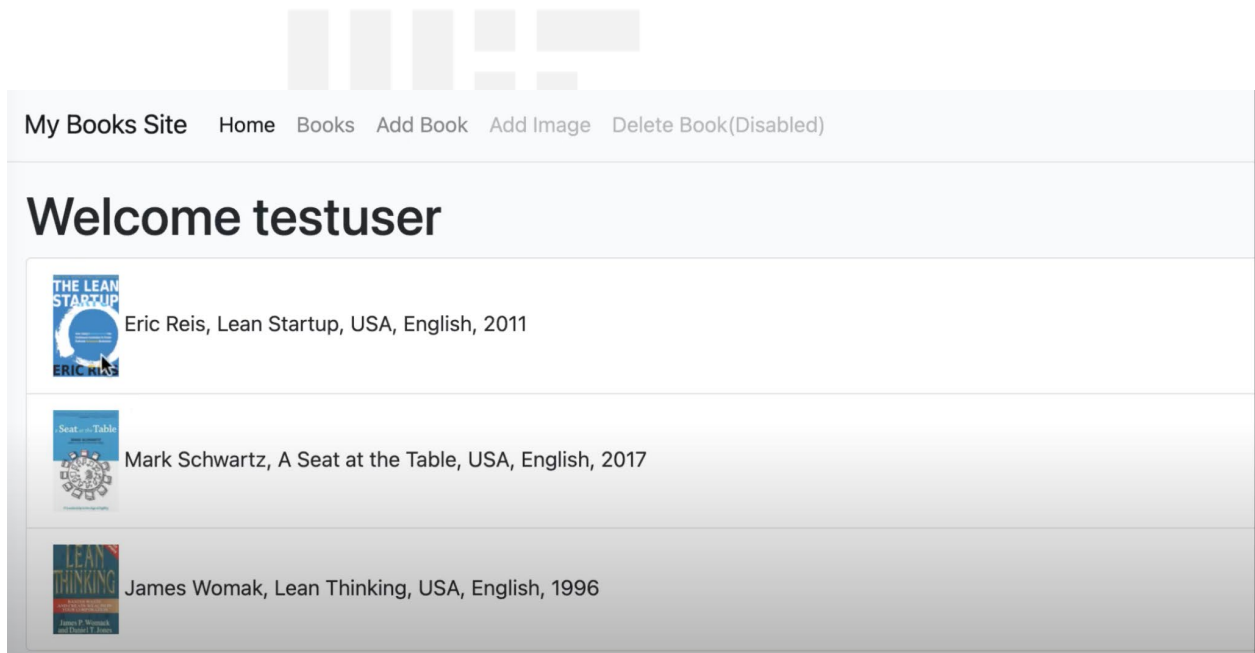
UserName

Password

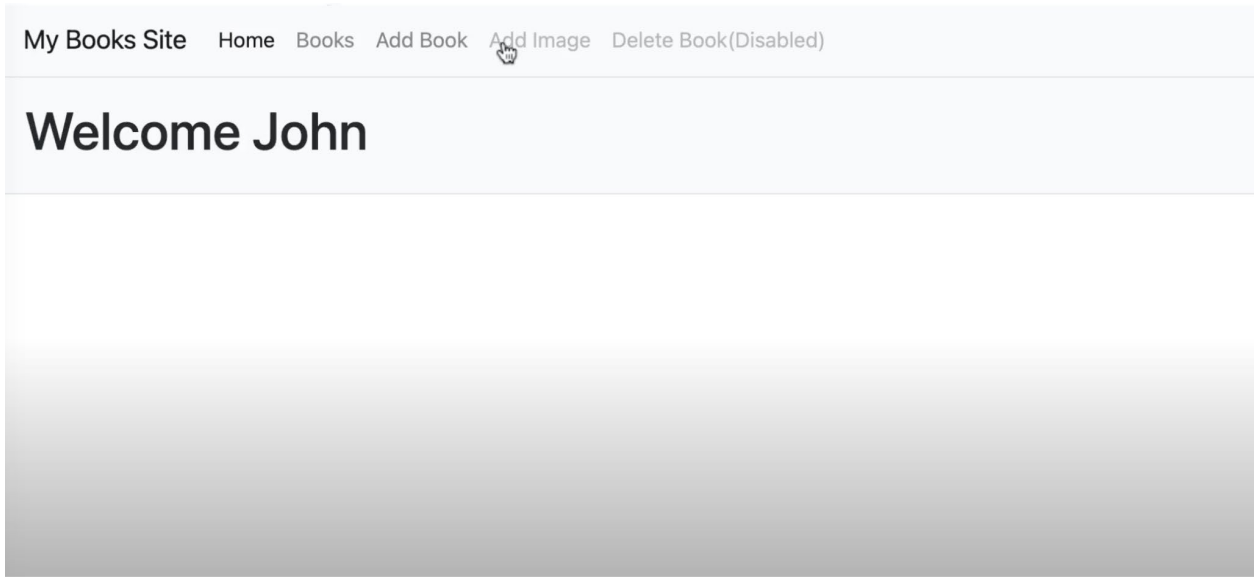
10.



11.



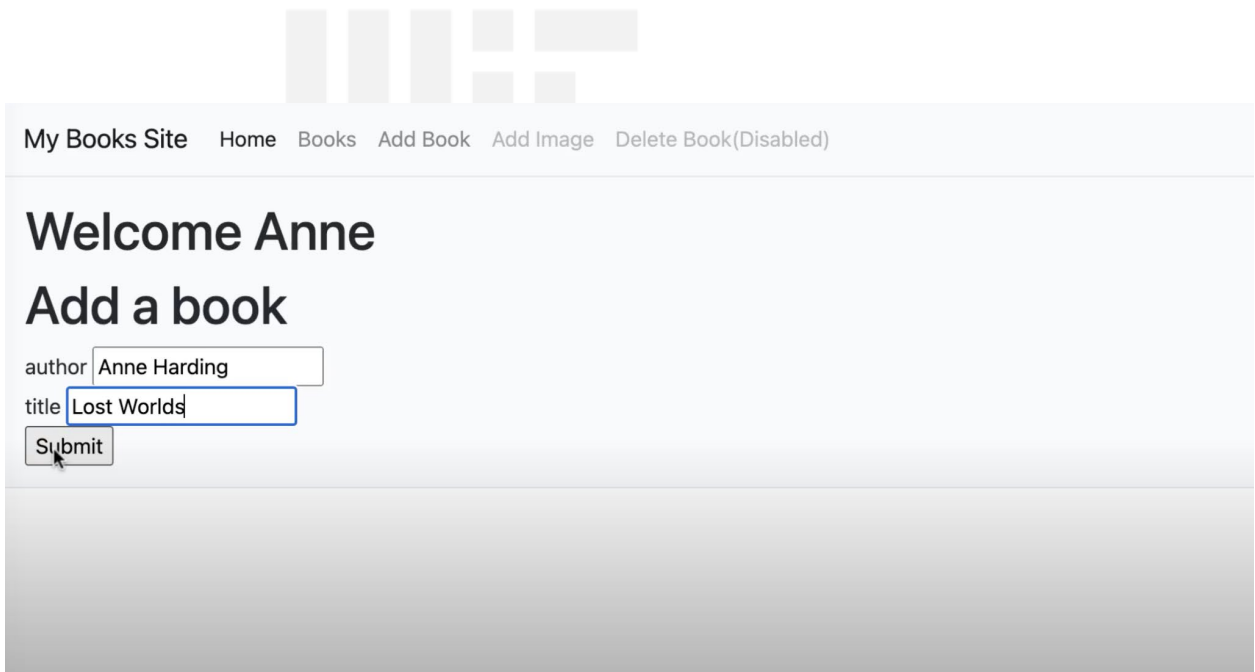
12.



My Books Site Home Books Add Book Add Image Delete Book(Disabled)

Welcome John

13.



My Books Site Home Books Add Book Add Image Delete Book(Disabled)

Welcome Anne

Add a book

author

title

14.

```
{  
  "msg": "Admins only!"  
}
```

Play (k)

